

Natural Orbitals : Theory and Applications

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I Theory on Natural Orbitals

This part of the notes is mainly inspired from [1, 2, 3].

Natural orbitals correspond to a change of basis of the naïve star representation of the Anderson impurity model (AIM). This transformation allows us to separate the bath into two parts: an active part and an inactive part, the latter being considered as a single Slater determinant and therefore becoming trivial for subsequent calculations (DMRG, ED, ...). Natural orbitals thus reduce the number of degrees of freedom to those associated with the active bath sites. In the MPS framework, this reduction can significantly reduce the required bond dimension.

Let us consider a one-band, spin-independent AIM (see also Fig. 1a):

$$\hat{H}_\star = U\hat{n}_{0\uparrow}\hat{n}_{0\downarrow} + \epsilon_0(\hat{n}_{0\uparrow} + \hat{n}_{0\downarrow}) + \sum_{l=1,\sigma}^{N_{\text{bath}}} \epsilon_l \hat{n}_{l\sigma} + V_l(\hat{c}_{0\sigma}^\dagger \hat{c}_{l\sigma} + h.c.) = U\hat{n}_{0\uparrow}\hat{n}_{0\downarrow} + \hat{H}_\star^{1B} \quad (1)$$

where site 0 corresponds to the impurity and $\hat{H}_\star^{1B}$ the one-body part of the AIM. We assume that we have access to the correlation matrix $C_{ij\sigma}$ (or density matrix) of the ground state $|\psi_{GS}\rangle$ of the AIM, calculated using DMRG:

$$C_{ij} = \langle \psi_{GS} | \hat{c}_i^\dagger \hat{c}_j | \psi_{GS} \rangle \quad (2)$$

where we omit the spin index. The correlation matrix can also be written in terms of its impurity and bath components:

$$C = \begin{pmatrix} \rho_{\text{imp}} & \rho_{ib} \\ \rho_{bi} & \rho_{\text{bath}} \end{pmatrix} \quad (3)$$

The next step consists of diagonalizing the bath part of the density matrix, ρ_{bath} . We do not diagonalize the full correlation matrix in order to avoid mixing the impurity site with the bath sites and therefore losing the locality of the impurity.

Because of the local interaction on the impurity, the new occupations are not strictly equal to 0 or 1. However, most of them remain very close to these values and can therefore be considered as part of a single Slater determinant as we will see later. Furthermore, from the properties of Gaussian states, there exists a single eigenvalue equal to $1 - \nu_0$, with $\nu_0 = \nu_{\text{imp}} = C_{00}$ being the impurity occupation, which is called the *active site* $|b\rangle$. In ascending order,

$$P_{\text{bath}}^\dagger \rho_{\text{bath}} P_{\text{bath}} = \text{diag}(\nu_1, \nu_2, \dots, \nu_{\text{active}}, \dots, \nu_{N_{\text{bath}}}) \quad (4)$$

where we consider the (almost) empty space $\mathcal{E}$ as $m \in \{1, \dots, N_{\text{active}} - 1\}$ and the (almost) filled space $\mathcal{F}$ as $n \in \{N_{\text{active}} + 1, \dots, N_{\text{bath}}\}$. Applying this change of basis gives (Fig. 1b)

$$\hat{H}' = P^\dagger \hat{H}_\star^{1B} P, \quad P = \begin{pmatrix} 1 & 0 \\ 0 & P_{\text{bath}} \end{pmatrix} \quad (5)$$

After this transformation, the impurity and active states are still strongly hybridized with all the empty and filled sites. However, the hopping between the empty $\mathcal{E}$ sites and the filled $\mathcal{F}$ sites is now almost zero (Fig. 1b). We therefore want to preserve the separation between the empty and filled states while further reducing the number of hybridization between the impurity and the conduction / valence bath (and same with the active site). This is necessary in order to construct an inactive space that is decoupled from the impurity and can therefore be treated trivially as a single Slater determinant. This can be achieved by performing a new unitary rotation within the subspace formed by the impurity and the active site. The goal of this rotation is to further separate $\hat{H}'$ into two *empty* and *filled* blocks. The resulting rotated states are called the *bonding* and *antibonding* states, with occupations approximately equal to 1 and 0, respectively:

$$|B\rangle = \frac{1}{\sqrt{2}} |\text{imp}\rangle + \frac{1}{\sqrt{2}} |b\rangle \quad (6)$$

$$|AB\rangle = \frac{1}{\sqrt{2}} |\text{imp}\rangle - \frac{1}{\sqrt{2}} |b\rangle \quad (7)$$

We finally reorder the sites such that the Hamiltonian takes a two-block form:

$$1, 2, \dots, N_{\text{active}} - 1, \text{antibonding}, \text{bonding}, N_{\text{active}} + 2, \dots, N_{\text{bath}} \quad (8)$$

where the two transformations (antibonding/bonding rotation and reordering) can be expressed by the following unitary matrix:

$$C = \begin{pmatrix} 0 & & & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & \\ 1 & 0 & & & & & & \\ & \ddots & \ddots & & & & & \\ & & 1 & 0 & & & & \\ & & & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & \\ & & & & & 1 & & \\ & & & & & & \ddots & \\ & & & & & & & 1 \end{pmatrix} \quad (9)$$

so that the new hamiltonian (Fig. 1c) can be expressed as

$$\hat{H}'' = C^\dagger P^\dagger \hat{H}_\star^{1B} P C = \begin{pmatrix} H_{\text{empty}} & 0 \\ 0 & H_{\text{filled}} \end{pmatrix} \quad (10)$$

Last step corresponds to tridiagonalize each block using Lanczos or Householder transformation, and then undo the transform C to recover the active and impurity site (keep locality !) (Fig. 1d)

$$\hat{H}_{NO}^{1B} = C Q^\dagger C^\dagger P^\dagger \hat{H}_\star^{1B} P C Q C^\dagger ; \quad Q = \begin{pmatrix} Q_{\text{empty}} & 0 \\ 0 & Q_{\text{filled}} \end{pmatrix} \quad (11)$$

Therefore, the natural orbital transformation constructs two chains, one corresponding to the (almost) empty states and the other to the (almost) filled states. The impurity and active sites, which remain coupled to each other, are "located" next to the first site of each chain and are connected

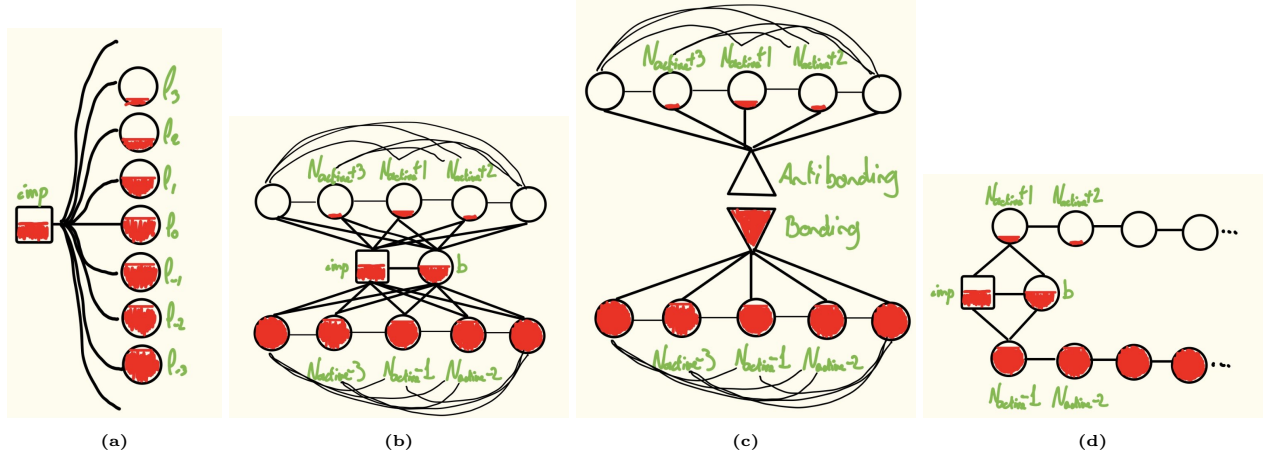


Figure 1: (a) Star representation of the AIM. (b) First rotation of $\hat{H}_*$ consisting of diagonalizing the bath density matrix. (c) Rotation of the impurity and active states to bonding and anti-bonding states with occupation of 1 and 0 respectively. (d) Hamiltonian in natural orbital basis after a tridiagonalization of the empty and filled parts and a rotation back to impurity and active states.

by a hopping term (Fig. 1d). The full natural-orbital Hamiltonian takes the form

$$\begin{aligned}
 \hat{H}_{NO} = & U \hat{n}_{0\uparrow} \hat{n}_{0\downarrow} + \epsilon_0 (\hat{n}_{0\uparrow} + \hat{n}_{0\downarrow}) + \sum_{\sigma} \left(V_{0N_{\text{active}}-1} \hat{c}_{0\sigma}^{\dagger} \hat{c}_{N_{\text{active}}-1\sigma} + V_{0N_{\text{active}}+1} \hat{c}_{0\sigma}^{\dagger} \hat{c}_{N_{\text{active}}+1\sigma} \right) \\
 & + \epsilon_b (\hat{n}_{b\uparrow} + \hat{n}_{b\downarrow}) + \sum_{\sigma} \left(V_{bN_{\text{active}}-1} \hat{c}_{b\sigma}^{\dagger} \hat{c}_{N_{\text{active}}-1\sigma} + V_{bN_{\text{active}}+1} \hat{c}_{b\sigma}^{\dagger} \hat{c}_{N_{\text{active}}+1\sigma} \right) + \sum_{\sigma} V_{0b} \hat{c}_{b\sigma}^{\dagger} \hat{c}_{0\sigma} \\
 & + \sum_{m=1,\sigma}^{N_{\text{active}}-1} \left(\epsilon_m \hat{n}_{m\sigma} + t_{mm+1} \hat{c}_{m\sigma}^{\dagger} \hat{c}_{m+1\sigma} \right) + \sum_{n=N_{\text{active}}+1,\sigma}^{N_{\text{bath}}} \left(\epsilon_n \hat{n}_{n\sigma} + t_{nn+1} \hat{c}_{n\sigma}^{\dagger} \hat{c}_{n+1\sigma} \right)
 \end{aligned}$$

For clarity, the complex conjugate of each off-diagonal term is not explicitly written. For the second line, the sum over the spin corresponds to the hopping between the impurity and the first site of the empty chain, with index $N_{\text{active}} - 1$ (resp. the first site of the filled chain, with index $N_{\text{active}} + 1$). For the third line, the first term corresponds to the on-site energy of the active site b . The second term describes the hopping between the empty and filled chains, while the last term corresponds to the hopping between the active site and the impurity. Finally, for the last line, the first term corresponds to the on-site energies of the filled chain and to the hopping between nearest-neighbor sites. The same applies to the second term for the empty chain. We can therefore rewrite the one-body matrix in a more compact form:

$$\hat{H}_{NO}^{1B} = \begin{pmatrix} \epsilon_0 & & & V_{0N_{\text{active}}-1} & V_{0b} & V_{0N_{\text{active}}+1} & & & \\ & \epsilon_1 & t_{12} & & & & & & \\ & t_{12} & \ddots & & & & & & \\ & & & \ddots & & & & & \\ V_{0N_{\text{active}}-1} & & & \epsilon_{N_{\text{active}}-1} & V_{N_{\text{active}}-1b} & & & & \\ V_{0b} & & & V_{N_{\text{active}}-1b} & \epsilon_b & V_{N_{\text{active}}+1b} & & & \\ V_{0N_{\text{active}}+1} & & & & V_{N_{\text{active}}+1b} & \epsilon_{N_{\text{active}}+1} & & & \\ & & & & & \ddots & & & \\ & & & & & \ddots & & & \\ & & & & & & t_{N_{\text{bath}}-1,N_{\text{bath}}} & & \\ & & & & & & t_{N_{\text{bath}}-1,N_{\text{bath}}} & \epsilon_{N_{\text{bath}}} \end{pmatrix} \quad (12)$$

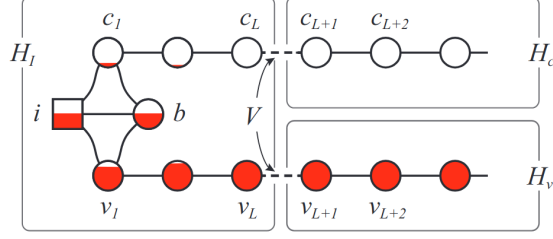


Figure 2: Separating the full Anderson impurity Hamiltonian $\hat{H}_{NO}$ defined on the natural orbitals into four parts $\hat{H}_I$, $\hat{H}_c$, $\hat{H}_v$, and V . Each Hamiltonian acts on the sites enclosed by its corresponding box. The hybridization operator V (dashed bonds) connects $\hat{H}_I$ to $\hat{H}_c$ and $\hat{H}_v$ [2].

I.1 Groundstate calculation

The computational complexity still grows exponentially with the number of bath sites. Let's recall the occupations of the bath sites after diagonalize the bath density matrix,

$$P_{\text{bath}}^\dagger \rho_{\text{bath}} P_{\text{bath}} = \text{diag}(\nu_1, \nu_2, \dots, \nu_{N_{\text{empty}}}, \dots, \nu_{\text{active}}, \dots, \nu_{N_{\text{filled}}}, \dots, \nu_{N_{\text{bath}}}) \quad (13)$$

where we define the *inactive empty bath part* as $i \in \{1, \dots, N_{\text{empty}}\}$, for which occupations ν_i of the empty chain is lower than ϵ , and the *inactive filled bath part* as $j \in \{N_{\text{filled}}, \dots, N_{\text{bath}}\}$, for which occupations ν_j of the full chain is greater than $1 - \epsilon$. The *active part* is thus defined by $k \in \{N_{\text{empty}} + 1, \dots, N_{\text{filled}} - 1\}$.

These parts correspond to the operator $\hat{H}_v$ in Fig. 2 for the inactive filled sites, with $L+1$ analogous to N_{filled} , and to the operator $\hat{H}_c$ in Fig. 2 for the inactive empty sites, with $L+1$ analogous to N_{empty} . Since the occupations of these orbitals are very close to 1 or 0, respectively, they can be accurately approximated as fully occupied or empty orbitals and therefore treated as a single Slater determinant. Truncating these inactive parts gives us access to the relevant part of the Hamiltonian, namely the active part represented by $\hat{H}_I$ in Fig. 2, which contains the non-trivial many-body physics of the system. In this way, the number of bath sites that needs to be explicitly treated in the many-body calculation can be reduced to only a few sites, with the required number depending on the chosen threshold ϵ and consequently on the desired accuracy of the calculation. For example, if we have access to the ground state $|\psi_{GS}^I\rangle$ of $\hat{H}_I$ (DMRG, ED), the total ground-state energy is given by

$$E_0(\epsilon) = \langle \psi_{GS}^I | \hat{H}_I | \psi_{GS}^I \rangle + \sum_{j=N_{\text{filled}}}^{N_{\text{bath}}} \epsilon_j \quad (14)$$

where the second term corresponds to the sum of the on-site energies of the filled states.

I.2 One Shot Natural Orbital Transformation

I.3 Natural Orbital Transformation at each Lanczos iteration

II Application on an one band model

II.1 Discretization of the bath and groundstate calculation

We test the natural orbital transformation on a one-band model of copper oxide $d_{x^2-y^2}$. We first perform an *ab initio* density functional theory (DFT) calculation and wannierize the $d_{x^2-y^2}$ orbitals.

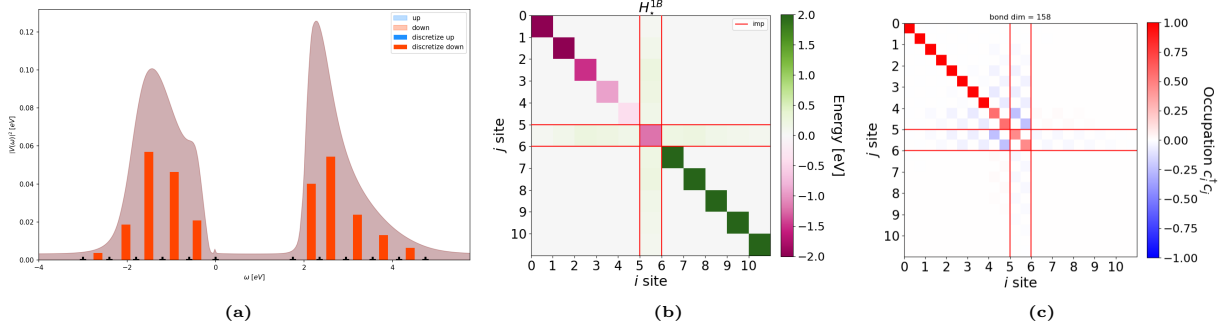


Figure 3: (a) Filled curve: continuous bath function in real frequency. Bars: discretization of the bath using 10 bath sites. (b) One-body part of the AIM for 10 bath sites (spin-independent) (1), with the impurity at the center (index 5), the conduction bath sites at indices 0 to 4, and the valence states at indices 6 to 10. The plot is shown with $vmin = -2$ and $vmax = 2$ to better visualize the hybridization. (c) Density matrix $C_{ij\sigma}$ (2) of the ground state calculated using DMRG with the initial state (15). Each site is represented by a 2×2 spin matrix. I put the bond dimension of the groundstate at the top of the figure.

We then compute the imaginary-frequency hybridization function within dynamical mean field theory (DMFT) using the continuous-time segmentation solver (CTSEG) of TRIQS, and analytically continue the hybridization function to real frequencies using the maximum entropy method (Max-ENT). We finally discretize the bath following [4] (see Fig. 3a). Note that the U parameter of the Hubbard model, with a value of 4 eV, is chosen such that it reproduces the experimental gap of CuO, which is approximately 1 – 2 eV (Fig. 3a). Fig. 3b shows the one-body part of the AIM for 10 bath sites (1), with the impurity in the middle at index 5, the conduction bath sites between indices 6 and 10, and the valence states between sites 0 and 4. Furthermore, I set $vmin = -2$ and $vmax = 2$ in the plot in order to better visualize the hybridization.

I calculate the ground state of the full Hamiltonian using DMRG. Interestingly, I encounter an error in the SyTen DMRG function when I use a random initial state with zero magnetization at half-filling. I therefore also consider an initial state in which the valence sites are filled, the conduction sites are empty, and the impurity spin is initialized in a superposition, such that

$$|\psi_{init}\rangle = \frac{1}{\sqrt{2}} |n_6 \uparrow = 1, n_6 \downarrow = 1, \dots, n_{10} \downarrow = 1, \mathbf{imp} \uparrow = \mathbf{1}, \mathbf{imp} \downarrow = \mathbf{0}, n_1 \uparrow = 0, \dots, n_5 \downarrow = 0\rangle \\ + \frac{1}{\sqrt{2}} |n_6 \uparrow = 1, n_6 \downarrow = 1, \dots, n_{10} \downarrow = 1, \mathbf{imp} \uparrow = \mathbf{0}, \mathbf{imp} \downarrow = \mathbf{1}, n_1 \uparrow = 0, \dots, n_5 \downarrow = 0\rangle$$

However, the ground states do not have zero magnetization: the impurity is either spin-up or spin-down. I therefore use a more "stable" initial state suggested by Sebastian, in which the correct symmetries (particle number and magnetization) are preserved for each Slater determinant. The drawback is that this initial state is not half-filled.

$$|\psi_{init}\rangle = \frac{1}{\sqrt{2}} |n_6 \uparrow = 1, \dots, \mathbf{n}_{10} \uparrow = \mathbf{1}, \mathbf{n}_{10} \downarrow = \mathbf{0}, \mathbf{imp} \uparrow = \mathbf{0}, \mathbf{imp} \downarrow = \mathbf{1}, n_1 \uparrow = 0, \dots\rangle \\ + \frac{1}{\sqrt{2}} |n_6 \uparrow = 1, \dots, \mathbf{n}_{10} \uparrow = \mathbf{0}, \mathbf{n}_{10} \downarrow = \mathbf{1}, \mathbf{imp} \uparrow = \mathbf{1}, \mathbf{imp} \downarrow = \mathbf{0}, n_1 \uparrow = 0, \dots\rangle \quad (15)$$

We show in Fig. 3c the density matrix $C_{ij\sigma}$ (2) of the ground state. Each site is represented by a 2×2 spin matrix. We first note that our system is indeed spin-independent, as expected since there is no spin interaction. Furthermore, the relevant symmetries are respected, with the correct total occupation and magnetization. Interestingly, the last valence site, site 4, has a slightly larger occupation (~ 1.1) than the impurity (~ 0.9), while both have zero magnetization. We can also observe non-zero off-diagonal terms involving the impurity and the last valence site, indicating that

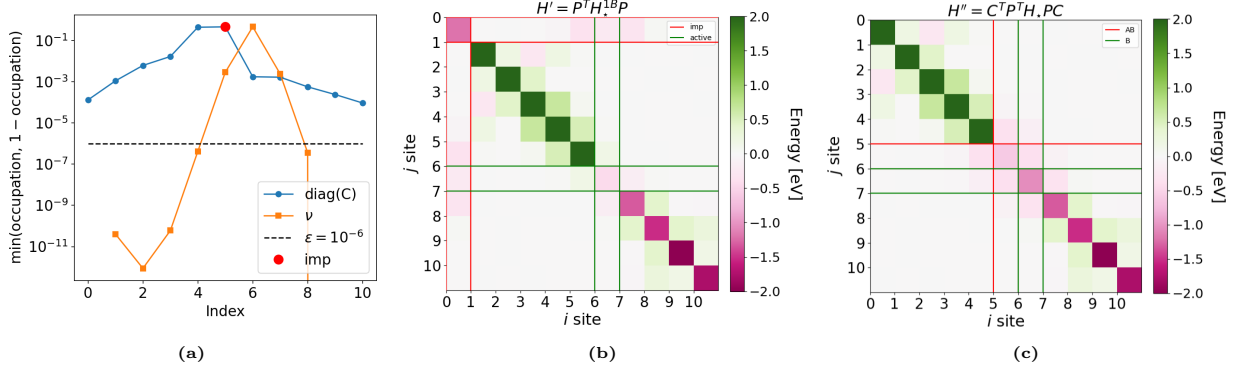


Figure 4: (a) Occupations of the different sites, showing how close they are to being empty or fully occupied (y-log scale). In blue: diagonal elements of the correlation matrix. In orange: eigenvalues of the bath density matrix ρ_{bath} . The threshold $\epsilon = 10^{-6}$ is also indicated as an example. (b) New Hamiltonian after the first rotation P , defined by the eigenvectors of ρ_{bath} , applied to the star representation of the AIM. The first site corresponds to the impurity, while the sixth site corresponds to the active site.

the system is non-trivial (in the sense that the occupation matrix is not just diagonal). In the following, we will work with spin up.

II.2 Natural Orbital Transformation

We first diagonalize the bath density matrix ρ_{bath} . In Fig. 4a, we show how close the occupations of the different bath orbitals are to 0 or 1. We compare, in blue, the diagonal elements of the correlation matrix with the impurity located at index 5, and in orange the eigenvalues of the bath density matrix. Note that the diagonalization mixes the bath states, so the site indices are no longer the same in the two representations. Moreover, the impurity site is not included in the bath space. As mentioned in the first chapter, some of the occupations are very close to 0 or 1 after diagonalization and can therefore be considered inactive. In our case, we choose a threshold of $\epsilon = 10^{-6}$, meaning that the first four bath sites are considered empty ($N_{\text{empty}} = 4$) and the last three are considered filled ($N_{\text{filled}} = 8$). The active space then consists of only three bath sites, with indices 5, 6, and 7. Furthermore, following the rule that the active site b must have an occupation ν_{active} equal to $1 - \nu_{\text{imp}}$, we can identify the state with index 6 as our active site, with an error of approximately $1 - \nu_{\text{active}} - \nu_{\text{imp}} \sim 10^{-3}$.

In Fig. 4b, we present the first rotation P defined by the eigenvectors of the bath density matrix (5) and applied to the hamiltonian. We place the impurity at index 0 to follow the convention introduced in the theoretical description. However, at the end of the natural orbital transformation, it may be preferable to place the impurity next to the active site in order to reduce entanglement at the MPS level. As expected from the theory, we obtain two blocks with all-to-all hoppings: the first corresponds to the (almost) empty space $\mathcal{E}$, with indices $m \in \{1, \dots, 5\}$, while the second corresponds to the (almost) filled space $\mathcal{F}$, with indices $n \in \{7, \dots, 10\}$. We can also observe a strong hybridization between the impurity and the active site, as well as between the impurity and the nearest neighbors of the active site.

In Fig. 4c, we show the second bonding/antibonding rotation C in Eq. 9 applied to the Hamiltonian $\hat{H}'$. Surprisingly, the resulting matrix is not clearly separated into two blocks as expected, with the separation occurring at site 6 : we observe some residual hybridization between the bonding and antibonding sites. Furthermore, when computing the density matrix restricted to these two states,

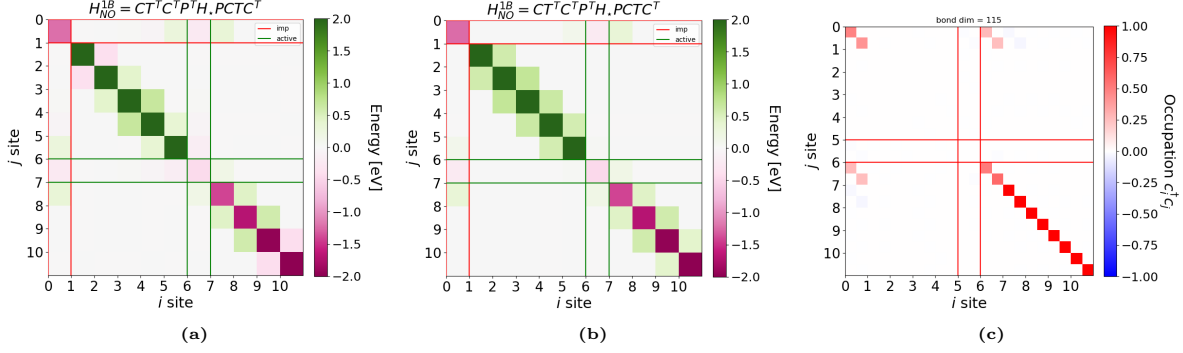


Figure 5: (a) One-body part of the natural-orbital Hamiltonian for the spin-up channel and (b) spin-down channel. (c) Density matrix in the ground state in the NO basis with 2×2 spin matrix per site (there is a mistake : the red lines should be at the site 0 corresponding to the impurity. The active site is still at site 6).

I obtain

$$\rho_{AB/B} = \begin{pmatrix} \langle AB|C|AB\rangle & \langle B|C|AB\rangle \\ \langle AB|C|A\rangle & \langle B|C|B\rangle \end{pmatrix} = \begin{pmatrix} 0.243 & -0.043 \\ -0.043 & 0.757 \end{pmatrix} \quad (16)$$

which gives occupations significantly different from 0 and 1 for the antibonding and bonding sites respectively, and non-zero off-diagonal terms. It is unclear whether this is a physical result, in which case I need to understand its origin and determine how to proceed with the subsequent steps, or whether it indicates that the rotation is not the appropriate one. In the latter case, which rotation matrix should be used instead? Any insight on this issue would be welcome.

I nevertheless proceed with the tridiagonalization of each block using a Householder transformation, and then transform back to the impurity-active basis. I show the resulting natural-orbital one-body Hamiltonian in Fig. 5a. It appears to exhibit the expected behavior, with hybridization between the impurity, the active site, and the conduction/valence sites (see (12)), with typical hopping amplitudes of order ~ 0.3 eV. Surprisingly, I do not obtain exactly the same off-diagonal matrix elements for the spin-down Hamiltonian (Fig. 5b). This is unexpected because the star AIM has exactly the same parameters for spin-up and spin-down, and the corresponding density matrices are also approximately identical, with differences of order $\sim 10^{-6}$. There are therefore several possible explanations: either there is an issue in my implementation, the Householder transformation is not unique (although I do not think this is the case), or the finite precision of the density matrix has a larger impact on the Householder transformation than expected. As an additional check, I compare the eigenvalues of $\hat{H}_*^{1B}$ and $\hat{H}_{NO}^{1B}$ and find differences of order 10^{-7} for both spin channels. This confirms that the implemented transformations preserve the eigenvalues to high accuracy and are therefore correctly unitary (it does not mean that it is the right transformation tho !). We show in Fig. 5c the density matrix in the ground state (with random initial state) in the NO basis with 2×2 spin matrix per site (there is a mistake : the red lines should be at the site 0 corresponding to the impurity. The active site is still at site 6). We have indeed empty sites between sites 1 and 4 and filled sites between sites 7 and 10. The impurity and active site are half-filled, with some correlation between them, and the impurity and the conduction / valence chains which are lower. I also check the ground-state energy of the full star Hamiltonian and that of the full natural-orbital Hamiltonian, including the U interaction. I obtain $E_0^* = -16.0500632$ eV and $E_0^{NO} = -16.0221970$ eV. This is encouraging, especially since I use a random initial state for the natural-orbital DMRG calculation. I should perform an additional calculation using an initial state similar to the one used for the star representation, with the valence states filled and the conduction states empty, in order to check whether the natural-orbital DMRG calculation converges to the correct ground-state minimum.

In Fig. 6, we show the natural-orbital transformation for AIMs with 11 bath sites (Figs. 6a, 6b, 6c), 13 bath sites (Figs. 6d, 6e, 6f), 16 bath sites (Fig. 6g, 6h, 6i), and 26 bath sites (Figs. 6j, 6k, 6l). The three columns show the successive steps of the transformation: (i) diagonalization of the bath density matrix followed by the bonding/antibonding rotation (Figs. 6a, 6d, 6g and 6j), (ii) tridiagonalization of the resulting empty and filled blocks (Figs. 6b, 6e, 6h and 6k), and (iii) the density matrix in the final natural-orbital basis using the groundstate calculated by DMRG (Figs. 6c, 6f, 6i and 6l). We also write in caption the difference between the groundstate energies of the star and natural orbital representations. Furthermore, the impurity is located at site 0, while the active site is located at sites 7, 8, 9, and 14, respectively (the red lines should be ignored).

In all cases, we still observe the problem of residual hybridization between the bonding and anti-bonding states in the first column (i). For the first and second rows, corresponding to 11 and 13 bath sites, respectively, we obtain the expected natural-orbital matrix pattern in the second column (ii) (Figs. 6b and 6e), in agreement with the theoretical form given in (12). However, the 11-bath-site system (Fig. 6b) exhibits an unusually large hybridization between the impurity–conduction chain and the active–conduction chain. This is also reflected in the density matrix (Fig. 6c), where the impurity is almost fully spin-up while the active site is almost fully spin-down. The empty chain is also not sufficiently empty (see site 6). Furthermore, we observe strong correlations between the active site and its second- and third-nearest valence neighbors (sites 8 and 9). The situation becomes even worse for the 16- and 26-bath-site systems: in the one-body natural-orbital Hamiltonian (ii) (Figs 6h and 6k), the impurity starts to hybridize with other sites of the conduction chain, whereas these terms are expected to vanish in the ideal natural-orbital representation. The density matrices obtained from the DMRG calculations (Figs 6i and 6l) are also not fully reliable, since these spurious hybridization terms are not included in the Hamiltonian used for the DMRG calculation as they should ideally be zero. The positive and relatively large difference between the ground-state energies of the star and natural-orbital representations for the 11-, 16-, and 26-bath-site systems (Figs. 6c, 6i and 6l) also indicates that something is not working properly.

These problems might again come from the fact that our hamiltonian is not well separated into empty and filled parts after the second bonding / anti-bonding rotation, and it seems to get worse and worse with the number of bath where the occupation of antibonding and bonding sites are getting closer to 1/2 each:

$$\rho_{AB/B}(N_{\text{bath}} = 11) = \begin{pmatrix} 0.243 & -0.043 \\ -0.043 & 0.756 \end{pmatrix}; \quad \rho_{AB/B}(N_{\text{bath}} = 13) = \begin{pmatrix} 0.274 & -0.032 \\ -0.032 & 0.725 \end{pmatrix} \quad (17)$$

$$\rho_{AB/B}(N_{\text{bath}} = 16) = \begin{pmatrix} 0.332 & -0.016 \\ -0.016 & 0.668 \end{pmatrix}; \quad \rho_{AB/B}(N_{\text{bath}} = 26) = \begin{pmatrix} 0.406 & -0.004 \\ -0.004 & 0.594 \end{pmatrix} \quad (18)$$

III Summary

The bonding/antibonding rotation does not produce the expected occupations of approximately one and zero: residual hybridization remains between these states. Despite this, the subsequent tridiagonalization gives a Hamiltonian with the expected overall structure, and the eigenvalues of the original and transformed one-body Hamiltonians agree within $\sim 10^{-7}$, confirming the numerical unitarity of the transformation. For larger baths, the transformation becomes less reliable. Spurious hybridizations appear in the conduction chain, while the expected block structure is only approximately recovered. The ground-state energies of the star and natural-orbital representations for the 10-bath-site system are nevertheless close, $E_0^* = -16.0501$ eV and $E_0^{\text{NO}} = -16.0222$ eV,

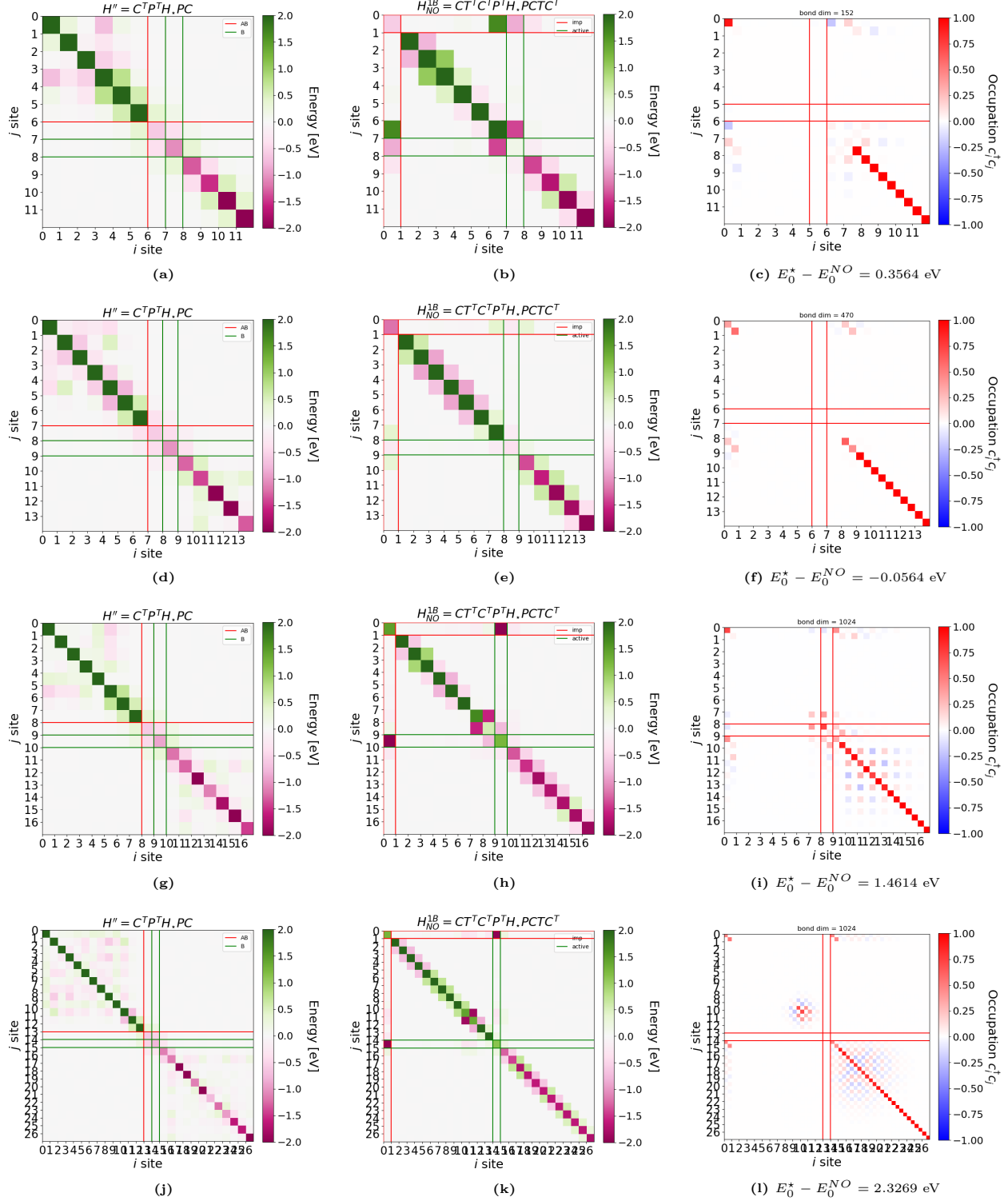


Figure 6: Natural-orbital transformation for AIMs with 11, 13, 16, and 26 bath sites (rows, from top to bottom). The three columns show the successive steps of the transformation: (i) diagonalization of the bath density matrix followed by the bonding/antibonding rotation, (ii) tridiagonalization of the resulting empty and filled blocks, and (iii) the density matrix in the final natural-orbital basis with the groundstate calculated with DMRG. We write in caption the difference between the groundstate energies of the star and natural orbital representations. Again for the last, the impurity index is at site 0 and active site at respectively 7, 8, 9 and 14 (ignore the red lines).

although further convergence tests are needed.

Overall, the results confirm that the natural-orbital transformation can identify the relevant active degrees of freedom, but the residual hybridizations and their worsening with bath size indicate that the current implementation still requires further investigation.

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